

Mohsen Faghihinezhad

CONTACT INFORMATION

A121 Research Engineering Complex,
Michigan State University, East Lansing, MI 48824
Phone: (517) 944-1673 | Email: Faghihi@msu.edu
Website: <https://mohsenfaghihinezhad.github.io/> | ORCID: [0000-0001-7180-3503](https://orcid.org/0000-0001-7180-3503)
Google Scholar: <https://scholar.google.com> | LinkedIn: <https://www.linkedin.com>

EDUCATION

Ph.D. in Environmental Engineering 2022-Present

Michigan State University, East Lansing, MI, USA | Expected Summer 2027 | GPA: 4.00/4.00

Advisor: Professor Alison M. Cupples

M.Sc. in Environmental Engineering 2018-2021

University of Tehran, Tehran, Iran | GPA: 3.5/4.0 (16.34/20)

Thesis: “Catalytic Ozonation of Real Textile Wastewater by g-C₃N₄ Modified with Al₂O₃ Nanoparticles”

Advisors: Professors M. Baghdadi and A. Torabian

B.Sc. in Civil Engineering 2013-2018

University of Qom, Qom, Iran | GPA: 2.63/4.0 (14.15/20)

RESEARCH INTERESTS

- Environmental microbiology and contaminant biotransformation
- Aerobic cometabolism and bioremediation of chlorinated ethenes and 1,4-dioxane
- Functional microbiomes, functional genes, and genome-resolved metagenomics
- Advanced oxidation and catalytic treatment processes
- Integrated biological and physicochemical treatment of contaminants of emerging concern
- Adsorption and water and wastewater treatment

RESEARCH EXPERIENCE

Graduate Research Assistant Sep. 2022-Present

Michigan State University, East Lansing, MI, USA

- Develop and characterize methanotrophic and propanotrophic enrichment cultures for the cometabolic biodegradation of chlorinated ethenes and 1,4-dioxane.
- Design and conduct batch biodegradation, substrate-utilization, and molecular experiments.
- Quantify functional genes using quantitative PCR and characterize microbial communities using sequencing-based methods.
- Perform genome-resolved metagenomic analyses and examine metagenome-assembled genomes and monooxygenase operons using KBase.

Graduate Research Assistant Sep. 2020-Sep. 2021

Nanotechnology Laboratory, University of Tehran, Tehran, Iran

- Investigated nanomaterials for water and wastewater treatment, including phosphate adsorption using an MgO-expanded graphite nanocomposite.
- Conducted catalytic ozonation experiments using industrial wastewater.

Research Assistant Sep. 2019-Sep. 2020

MNR Wastewater Treatment Plant Design, University of Tehran, Tehran, Iran

- Developed a decentralized municipal wastewater treatment approach and a pilot-scale model of the MNR treatment system.

PUBLICATIONS

Peer-Reviewed Journal Articles

- **M. Faghihinezhad**, Z. Eshghdoostkhatami, A.M. Cupples, Characterization of multiple trichloroethene, cis-dichloroethene and 1,1-dichloroethene degrading propanotrophic communities, *J. Environ. Manage.* 408 (2026) 129957. <https://doi.org/10.1016/J.JENVMAN.2026.129957>.
- **M. Faghihinezhad**, Z. Eshghdoostkhatami, A. Bernstein, A.M. Cupples, Identification of the dominant methanotrophs in trichloroethene degrading enrichment cultures from multiple sources, *J. Hazard. Mater.* 499 (2025) 140268. <https://doi.org/10.1016/J.JHAZMAT.2025.140268>.
- Z. Eshghdoostkhatami, Z. Li, **M. Faghihinezhad**, A.M. Cupples, Characterization of propanotrophic enrichments from agricultural soils capable of 1,4-dioxane biodegradation to sub- $\mu\text{g/L}$ levels, *Science of the Total Environment* 1005 (2025) 180824. <https://doi.org/10.1016/J.SCITOTENV.2025.180824>.
- M.M. Masoumi Sangani, M.S. Shahin, M.A. Yavari, **M. Faghihinezhad**, M. Baghdadi, Tailoring photocatalytic activity of graphitic carbon nitride using sulfonic acid and incorporating in chitosan beads: Cr(VI) removal from aqueous solutions, *J. Ind. Eng. Chem.* (2023). <https://doi.org/10.1016/J.JIEC.2023.09.047>.
- **M. Faghihinezhad**, M. Baghdadi, M.S. Shahin, A. Torabian, Catalytic ozonation of real textile wastewater by magnetic oxidized g-C₃N₄ modified with Al₂O₃ nanoparticles as a novel catalyst, *Sep. Purif. Technol.* 283 (2022) 120208. <https://doi.org/10.1016/J.SEPPUR.2021.120208>.
- M. Shirazinezhad, **M. Faghihinezhad**, M. Baghdadi, M. Ghanbari, Phosphate removal from municipal effluent by a porous MgO-expanded graphite composite as a novel adsorbent: Evaluation of seawater as a natural source of magnesium ions, *J. Water Process Eng.* 43 (2021) 102232. <https://doi.org/10.1016/J.JWPE.2021.102232>.
- B.H. Ali, M.S. Shahin, M.M.M. Sangani, **M. Faghihinezhad**, M. Baghdadi, Wastewater Aerosols Produced during Flushing Toilets, WWTPs, and Irrigation with Reclaimed Municipal Wastewater as Indirect Exposure to SARS-CoV-2, *J. Environ. Chem. Eng.* (2021) 106201. <https://doi.org/10.1016/j.jece.2021.106201>.

Book Chapters

- **M. Faghihinezhad**, Reclaimed wastewater in the food industry: Sources, treatment, and reuse. In: *Wastewater and Sludge Treatment for Reuse*, edited by S. K. Brar, H. Ratnaweera, and R. K. Das. Springer, 2025. https://doi.org/10.1007/978-93-985-1249-1_1249.

Conference Papers

- **M. Faghihinezhad**, M. Baghdadi, M.S. Shahin, A. Torabian. Catalytic ozonation of textile wastewater by g-C₃N₄ modified with Al₂O₃ nanoparticles (1st National Water Quality Management Conference) (November 2021).
- M. Shirazinezhad, **M. Faghihinezhad**, M. Baghdadi. Phosphate removal from municipal effluent by a porous MgO-expanded graphite composite as a novel composite (1st National Water Quality Management Conference) (November 2021).

CONFERENCE PRESENTATIONS AND POSTERS

- **M. Faghihinezhad**, Z. Eshghdoostkhatami, and A. M. Cupples. 2026. Characterization of Trichloroethene, cis-Dichloroethene, 1,1-Dichloroethene and 1,4-Dioxane Degrading Propanotrophic Enrichment Cultures (*Poster presentation*), 2026 Chlorinated Conference, Fort Worth, TX, May 31-June 4, 2026 (Battelle).

- **M. Faghihinezhad**, Z. Eshghdoostkhatami, and A. M. Cupples. 2026. Propanotrophic Enrichment Cultures Capable of the Biodegradation of Multiple Groundwater Contaminants (*Poster presentation*), 2026 Borchardt Conference: 27th Triennial Symposium on Advancements in Water & Wastewater, University of Michigan, Ann Arbor, MI, May 5–6, 2026.
- **M. Faghihinezhad**, Z. Eshghdoostkhatami, and A. M. Cupples. 2026. Propanotrophic Enrichment Cultures Expressing Propane Monooxygenase for the Cometabolic Biodegradation of Common Groundwater Contaminants (*Poster presentation*), CEE 2026 Symposium, Michigan State University, East Lansing, MI, 2026.
- **M. Faghihinezhad** and A. M. Cupples. 2025. Examining the Diversity of Methane Monooxygenases and Methanotrophs in Trichloroethene Degrading Cultures (*Platform presentation*), *Seventh International Symposium on Bioremediation and Environmental Biotechnology (Battelle)*, Boston, MA, June 2025.
- **M. Faghihinezhad** and A. M. Cupples. 2025. Methanotrophs with Novel Ammonium/Particulate Methane Monooxygenase Operons in Trichloroethene Degrading Microcosms (*Poster presentation*), *MSU College of Engineering Graduate Research Symposium*, East Lansing, MI, March 2025.
- **M. Faghihinezhad** and A. M. Cupples. 2024. Exploring Substrate Amendments for Trichloroethene Biodegradation in Soils and Sediments (*Poster presentation*), *ESPP 2024 Fall Student Research Symposium, Kellogg Hotel and Conference Center, Michigan State University*, East Lansing, MI, October 11, 2024.
- **M. Faghihinezhad** and A. M. Cupples. 2024. Identifying the Microorganisms Involved in Carbon Uptake from Trichloroethene under Different Incubation Conditions (*Poster presentation*), *2024 Chlorinated Conference*, Denver, CO, June 2-4, 2024.
- **M. Faghihinezhad** and A. M. Cupples. 2024. Examining the Impact of Propane and Methane on Trichloroethene Biodegradation in Soils and Sediments (*Poster presentation*), *2024 Engineering Graduate Research Symposium, Michigan State University*, East Lansing, MI, May 2024.

RESEARCH METHODS AND TECHNICAL SKILLS

Experimental methods: Microbial enrichment cultivation; batch biodegradation and substrate-utilization experiments; quantitative PCR.

Bioinformatics: Genome-resolved metagenomics; metagenome-assembled genomes; functional-gene and monooxygenase-operon analysis using KBase.

Programming and analysis: C++, MATLAB, R (RStudio).

Other software: Microsoft Excel, Word, PowerPoint, Mendeley, WaterGEMS, WAVE, AutoCAD, Civil 3D, ETABS.

SELECTED ACADEMIC PROJECTS

- Characterized wastewater from an antibiotic-manufacturing facility and designed an appropriate treatment system.
- Designed a municipal wastewater treatment plant and specified the individual treatment units.
- Evaluated secondary wastewater treatment performance for removing contaminants of emerging concern.

HONORS AND AWARDS

- KBS LTER Graduate Fellowship, Michigan State University, 2025.
- Elsevier Recognized Reviewer Certificates, 2021 and 2024.
- Ranked 704th among approximately 40,000 examinees in Iran's national master's entrance examination, 2018.
- Ranked in the top 1.5% of examinees in Iran's national university entrance examination, 2013.

TEACHING EXPERIENCE

Graduate Teaching Assistant

Aug. 2026-Present

Michigan State University, East Lansing, MI, USA

- Support course instruction, grading, problem-solving sessions, and student learning in ENV/ENE 380.
- Develop worked-example instructional videos for ENV/ENE 380, explaining environmental engineering calculations and problem-solving methods.

Graduate Teaching Assistant

Aug. 2025-Jan. 2026

Michigan State University, East Lansing, MI, USA

- Assisted with course instruction, grading, and student support for ENE 381 (Environmental Chemistry).

Teaching Assistant

Oct. 2020-Feb. 2021

University of Tehran, Tehran, Iran

- Assisted Professor Majid Baghdadi with instruction in the Water and Wastewater Laboratory course.

TEACHING CERTIFICATIONS AND PROFESSIONAL DEVELOPMENT

- Certification in College Teaching (CCT), Michigan State University (in progress).
- Colleges Online Learning Academy (COLA) Fellow, Michigan State University, 2026.

SERVICE AND OUTREACH ACTIVITIES

- Departmental Lab Tour Volunteer, Michigan State University, April 2025. Led environmental engineering research laboratory tours through the Service Learning and Civic Engagement initiative.
- Poster Judge, University Undergraduate Research and Arts Forum, Michigan State University, April 2024. Evaluated Health Sciences posters and provided feedback to presenters.
- Department Representative, MSU College of Engineering Graduate Recruitment Event, August 2023. Introduced prospective students to departmental research and graduate programs.

ADDITIONAL PROFESSIONAL EXPERIENCE

Graduate Student Assistant, Battelle. Supported technical teams and conference presentations.

- Chlorinated Solvents Conference, Fort Worth, TX, June 2026.
- Bioremediation Symposium, Boston, MA, June 2025.
- Bioremediation Symposium, Austin, TX, May 2023.

Construction Supervisor. Amin Complex, Qom, Iran (2016-2018). Supervised structural detailing and finishing work; supported cost estimation and construction-cost management.

REFERENCES

Professor Alison M. Cupples | Michigan State University
cupplea@msu.edu | (517) 432-3370

Professor Syed A. Hashsham | Michigan State University
hashsham@egr.msu.edu | (517) 355-8241

Dr. Kaisen Lin | Michigan State University
kaisen@msu.edu | (517) 884-8460

Dr. Majid Baghdadi | University of Tehran
m.baghdadi@ut.ac.ir | (+98) 21-6111 ext. 2778